

RASHTRIYA KISHOR SWASTHYA KARYAKRAM

API DOCUMENT for Peer Educator (PE) Data Exchange

Version 1.0

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Overview

Rastriya Kishor Swasthya Karyakram (RKSK) focuses on adolescent participation and leadership, Equity and inclusion, Gender Equity and strategic partnerships with other sectors and stakeholders. The website rksk.in offers an integrated ICT platform to all of its stakeholder to enhance the overall efficacy of the program.

This document highlights on the functional specification of integrating the work being done by the Peer Educator (PE). Based on the Key Performance Indicators (KPI), the data will be pulled from the server having the details of PE's activity.

Functional Specification

In the current IT System, the following KPIs are being collected

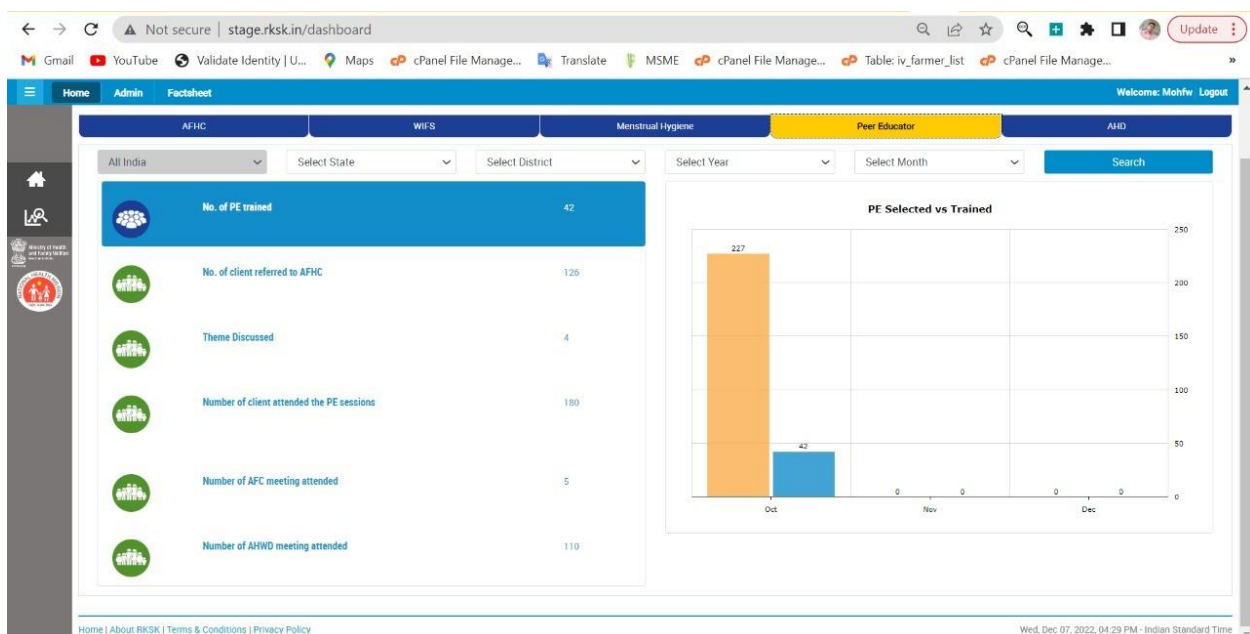
- No of PE Trained
- No of Client referred to AFHC

The new KPI identified will be visually presented on the PE section of the RKSK Dashboard. Based on the data schema (Reference Annex – I), the following KPIs may be considered

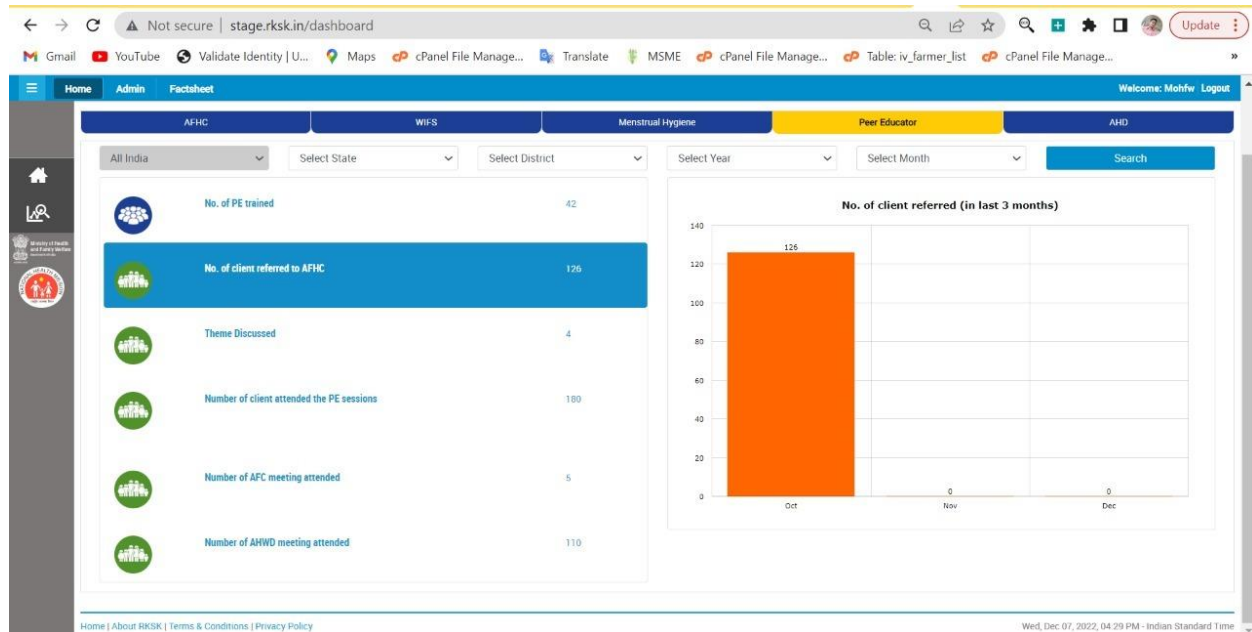
- Number of PE sessions conducted
- Number of adolescents attended the PE Session
- Thematic Coverage
- No of AFC meeting attended
- No of AHWD meeting attended

The proposed visual designs for the new indicators are given below:

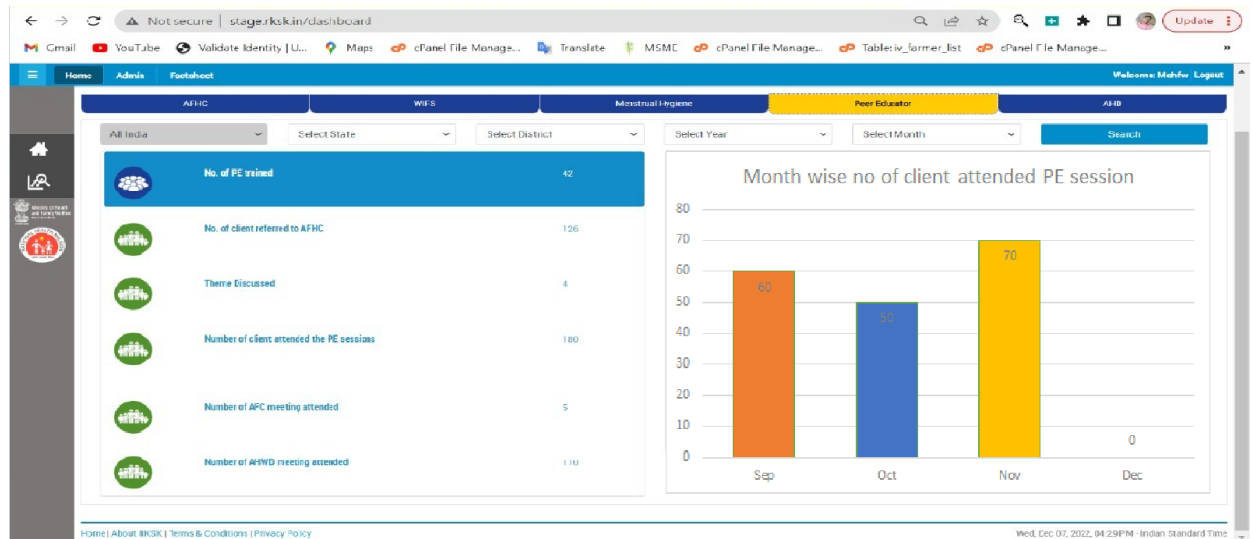
No. of PE Trained



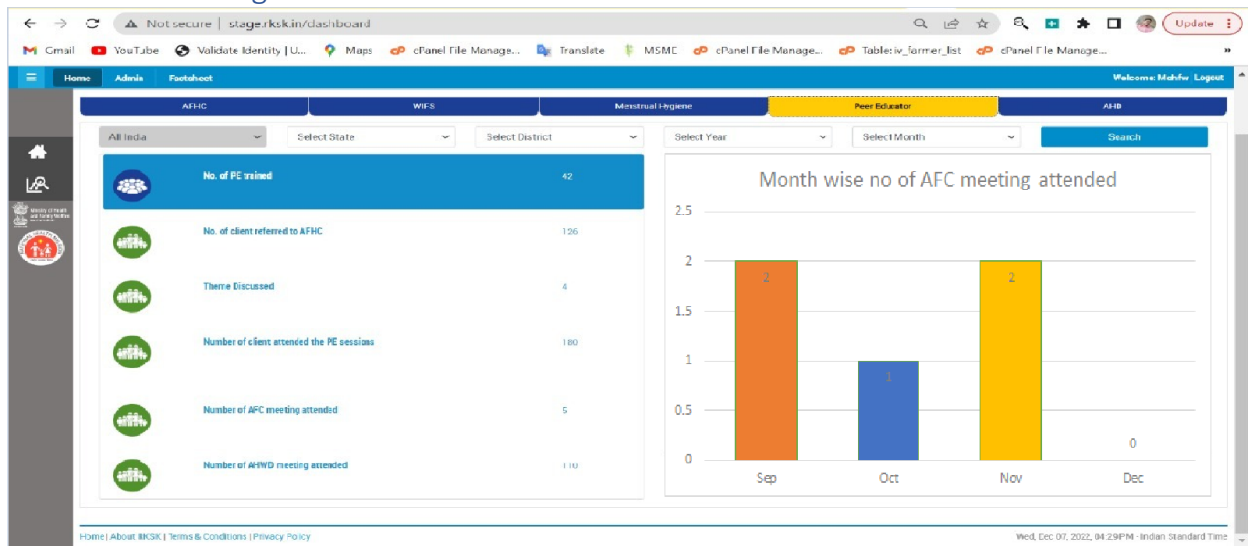
No. of Client referred to AFHC



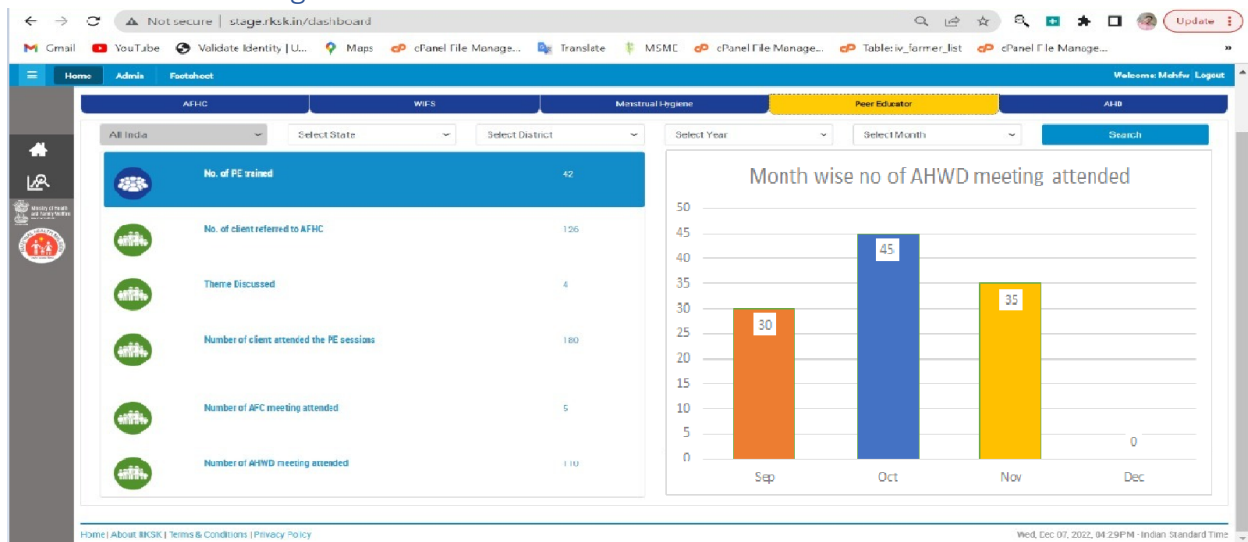
Number of clients attended the PE Session



No of AFC meeting attended



No of AHWD meeting attended



Assumption

The following compliances are required to enable the data exchange between the systems

Compliance to LGD (LOCAL GOVERNMENT DIRECTORY) code

Introduction Local Government Directory (LGD) is one of the applications developed as part of Panchayat Enterprise Suite (PES) under e-Panchayat Mission Mode project (MMP). LGD aims to keep all information about the structure of Local Governments and revenue entities online. Main objective of LGD is to maintain up-to-date list of revenue entities (districts/subdistricts/villages), Local Government Bodies (Panchayats, Municipalities and traditional bodies) and their wards, organizational structure of Central, State and Local Governments, reporting hierarchy within the government organizations and parliamentary and assembly constituencies and their relationship with one another. More about the same may be seen at <https://lgdirectory.gov.in/>.

All states, Districts, Blocks and any other locational entity must be fetched with the LGD code to avoid any mismatch.

API Queue Management

While enabling data exchange, it is important to have a Queue Management framework to take care of the load factor. This will avoid any error in the data exchange process. Appropriate error code must be given in all instances. The standard codes are given below:

403	{"error":"API key is missing."}
400	{"error":"Please provide valid username and password."}
401	{"error":"Invalid API key."}
401	{"error":"Incorrect username or password."}
500	{"error":"Something went wrong. Please try again later."}
200	Success Got Result
100	Result not available

Any custom response or change in the above-mentioned code must be notified.

Validation Protocol

To develop a secure system of data exchange, it is important to have well-defined protocol. The suggested steps are as follows

1. The calling system to send a token request message with the username and the password to the target API Server
2. Upon successful authentication, the target server would give a JWT token with a validity of predefined time. It is recommended to keep it 30 second.
3. The API to pull the data must be sent with the JWT Token. Without the JWT Token, the request should be ignored.
4. The API to send the desired data with appropriate code.
5. In case of any error, the defined return code must be returned

API Specification in Details

API: Get_Token

API Description: To get JWT Token from the target server

API Name	GetToken
HTTP Method	POST
Content-Type/ Response Format	Application/JSON
Input Parameter(s)	Username (Char), Password (Char)
Output Parameter	JWT Token ¹ (Example given below)
Response Code	
Response Format	JSON

"token": "eyJ0eXAiOiJKV1QiLCJhbGciOiJSUzI1NiJ9.eyJpc3MiOiJmaXJlYmFZS51hZG1pbmNkay10cWRqNkBjbGlTYXRlaGVybzV2LTkyZxcxLmlhbm55nc2VydmllZWJfYjY21bnQuYjY2tlliwc3ViljoIZmlyZWJhc2UtYWRTaW5zZGstdHFKajAY2xpbfWF0ZWhlcmlcyO5M2M3MS5pYW0uZ3NlcnZpY2VhY2NvdW50LmNvbSlslmF1ZC16lmh0dBzOlwvXC9pZGVudGI0eXRvb2xraXQxZ29vZ2xlYXBpcy5jb21c1L2dvb2dsZS5pZGVudGI0eS5pZGVudGI0eXRvb2xraXQudjEuSWRlbmRpdHUlb29sa2I0IiwiaWF0IjoxNjcwMzMwNzlyLCJleHAiOiE2NzAzMzQzMjlsInVpZC16IiwiaWF0IjE1LCJkbGpfbXMiOnsiZnVsbnBfY2VudW51LjoiOiJ0eXZGZDcj9fQ.Ig4oNQVvy26Tfx8xs2jjvtKFLBQApyS0rV1AsY3U39tntBOD3ICH8k2GvZ-C"

¹ JSON Web Token (JWT) is an open standard (RFC 7519) that defines a compact and self-contained way for securely transmitting information between parties as a JSON object. This information can be verified and trusted because it is digitally signed. JWTs can be signed using a secret (with the HMAC algorithm) or a public/private key pair using RSA or ECDSA. Although JWTs can be encrypted to also provide secrecy between parties, we will focus on signed tokens. Signed tokens can verify the integrity of the claims contained within it, while encrypted tokens hide those claims from other parties. When tokens are signed using public/private key pairs, the signature also certifies that only the party holding the private key is the one that signed it.

API: Get_State

API Description: To gets list of All States and UT Master with LGD Code

API Name	Download State
HTTP Method	POST
Content-Type/ Response Format	Application/JSON
Authorization	Basic Authentication with username and password Username: demo Password: demo@123
Input Parameter(s)	Nothing
Output Parameter	State_lgd_code(int), state_name(char)
Response Code	To be given based on the server response
Response Format	JSON

API: Get_District

API Description: To get list of all district based on State LGD Code

API Name	Download District
HTTP Method	POST
Content-Type/ Response Format	Application/JSON
Authorization	Basic Authentication with username and password Username: demo Password: demo@123
Input Parameter(s)	State_lgd_code (int)
Output Parameter	district_lgd_code(int), district_name(char), state_lgd_code(int)
Response Code	To be given based on the server response
Response Format	JSON

API: Get_Block

API Description: To gets list of all block based on district LGD code

API Name	Download Block
HTTP Method	POST
Content-Type/ Response Format	Application/JSON
Authorization	Basic Authentication with username and password Username: demo Password: demo@123
Input Parameter(s)	District_LGD_code (int)
Output Parameter	block_lgd_code(int), block_name(char), state_lgd_code(int), district_lgd_code(int)
Response Code	To be given based on the server response
Response Format	JSON

API: Get_PE_Activity_Data

API Description: To get the fields attributing PE's activity

HTTP Method	POST
Content-Type/ Response Format	Application/JSON
Authorization	Basic Authentication with username and password Username: demo Password: demo@123
Input Parameter(s)	State_lgd_code (int), from_date (date), to_date(date), token(jwt_token)
Output Parameter	PE_ID PE_name(char) from_date(date) to_date(date) no_of_adolescent_attended(int), no_of_adolescent_referr(int) afc_meeting(char), no_of_afc_meeting_attended(int) ahwd_meeting(char), no_of_ahwd_meeting_attended(int) state_lgd_code(int) district_lgd_code(int) block_lgd_code(int), theme_for_PE_session(char[])
Response Code	
Response Format	JSON